

MoReCMT: Multimodal Recurrent Cross-Modal Transformer-Based Feature Fusion for Emotional Recognition

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Project Repository: <https://github.com/ajha7/MemoCMTDialogueRNN>

Abstract

Cross-modal transformers like MemoCMT have significantly improved Multimodal Emotion Recognition (MER) by fusing audio and textual cues. However, they often treat utterances as isolated events. This "context blindness" ignores the sequential nature of human conversations, where we derive meaning from past context. To address this, I propose a hybrid architecture that uses MemoCMT as a dense feature extractor for a DialogueRNN, combining deep multimodal fusion with temporal context modeling. Experiments on the IEMOCAP dataset demonstrate that this combination yields quantifiable gains: a context window of 6 past utterances achieved an accuracy of 81.52%, outperforming the isolated baseline and confirming that historical context is essential for distinguishing emotional states.

1 Introduction

Human conversations are sequential. Take this example:

Turn 1 (Speaker A): "I got the promotion!" → *Joy*

Turn 2 (Speaker B): "That's the job I wanted. . ." → *Sadness*

Turn 3 (Speaker A): "We should celebrate!" → *Joy*

Turn 4 (Speaker B): "Sure." → ???

Interpreting a shorter, ambiguous phrase like "Sure" requires knowing what happened in the conversation. Was it excited or sad? Models like MemoCMT (Khan et al., 2025) are great at extracting the snapshot emotion of a clip but discard the trajectory of the dialogue. Older models like DialogueRNN (Majumder et al., 2019) were made to track speaker

states over time but they use much simpler feature extractors that miss the cross-modal nuances between text and audio.

This work tries to combine both papers. By feeding the fused embeddings from MemoCMT into the temporal tracking mechanisms of DialogueRNN, I aim to create a system that benefits from both the deeper feature extraction and the conversational memory. The core question here is whether giving a powerful transformer a "memory" yields practical performance gains.

This paper presents:

- Hybrid architecture utilizing MemoCMT embeddings to drive a windowed DialogueRNN
- Analysis of different context windows ($w = 1, 3, 6$) to quantify the value of history
- Ablation studies isolating the contributions of text vs. audio modalities

2 Related Work

Multimodal fusion typically falls into early, late, and hybrid approaches (Wu et al., 2025). Hybrid models like MemoCMT represent the state-of-the-art in multimodal audio/text fusion in dyadic conversations by (Khan et al., 2025) dynamically aligning audio and text. They currently operate on single utterances instead of including the temporal aspect of conversations.

To handle conversation dynamics, Majumder et al. (2019) introduced DialogueRNN, which uses GRUs to track speaker, global, and emotion states. Their work demonstrated that a speaker's emotion is a function of both their history and their partner's influence. The proposed approach simply swaps their feature extraction for the MemoCMT fusion model in hopes that the higher-quality inputs will improve the outputs from the temporal tracking.

3 Methods

3.1 Dataset

I utilized the IEMOCAP dataset, a standard benchmark of roughly 12 hours of dyadic sessions. The study focuses on audio and text modalities across four primary classes: happy, angry, sad, and neutral.

3.2 System Architecture

The pipeline can be broken into two stages: the raw audio/text data processed by the feature extractor(MemoCMT) to create dense embeddings, which is then passed to the context model (DialogueRNN).

3.2.1 Phase 1: Feature Extraction

I used the publicly available MemoCMT architecture which employed the pre-trained BERT model to process aligned text tokens and audio spectrograms. Instead of using the final classification, I extracted the fused feature vector z_t from the dense layer before the head. This preserves the rich, cross-modal information without collapsing it into a probability distribution.

3.2.2 Phase 2: Contextual Modeling

To "remember" the conversation, I implemented a sliding-window DialogueRNN that tracks 3 states over the given window w .

Global State (g_t): Captures the broad context via a Global GRU:

$$g_t = G_{global}(z_t, g_{t-1}, q_{r_{t-1}, t-1}) \quad (1)$$

Here, $q_{r_{t-1}, t-1}$ represents the interlocutor's(the other participant in the conversation) previous state.

Party State ($q_{s_t, t}$): Tracks the specific emotional trajectory of participant s :

$$q_{s_t, t} = G_{party}(z_t, g_{t-1}, q_{s_t, t-1}) \quad (2)$$

Emotion State (e_t): The final representation used for classification, derived from both party and global states:

$$e_t = \text{ReLU}(W_e[q_{s_t, t} \oplus g_t] + b_e) \quad (3)$$

This vector e_t is passed through a softmax layer to predict the label $\hat{y}_t$.

Model	UA	WA	W-F1
<i>Context</i>			
No context ($w = 1$)	76.2 $\pm$ 1.0	75.7 $\pm$ 0.8	75.5 $\pm$ 0.9
$w = 3$	80.3 $\pm$ 0.9	79.9 $\pm$ 0.7	79.8 $\pm$ 0.8
$w = 6$	81.5 $\pm$ 0.8	81.3 $\pm$ 0.6	81.4 $\pm$ 0.7
Whole dialogue	80.4 $\pm$ 1.4	80.2 $\pm$ 1.1	80.0 $\pm$ 1.2
<i>Modality (best $w = 6$)</i>			
Text only	68.5 $\pm$ 1.1	67.8 $\pm$ 0.9	67.4 $\pm$ 1.0
Audio only	68.9 $\pm$ 1.8	68.2 $\pm$ 1.5	67.6 $\pm$ 1.7
Plain MemoCMT, same split	75.2 $\pm$ 1.1	75.7 $\pm$ 0.9	75.5 $\pm$ 1.0

Table 1: Results on the IEMOCAP test set (Session 5), reported as mean $\pm$ standard deviation over three seeds. UA: unweighted accuracy; WA: weighted accuracy; W-F1: weighted F1. Best results are in bold.

3.3 Implementation Details

The training used weighted cross-entropy loss in order to handle the class imbalance. I used the AdamW optimizer, the MemoCMT encoders were frozen (or updated in the last layers at 5×10^{-6} and a couple other learning rates), while the DialogueRNN components were trained at 1×10^{-4} .

Note: some experiments on fine-tuning different layers are missing from the open sourced materials

4 Results and Discussion

The primary experiment measured the effect of changing the context window w and how it affected validation accuracy.

I could not perfectly replicate the original MemoCMT paper's reported accuracy (81.33%) due to the unavailability of their specific BERT fine-tuning. The *relative* improvement provided by the architecture is what is important here.

4.1 The Impact of Context

There is a clear improvement as the context window expands. Moving from a single utterance ($w = 1$) to a short history ($w = 3$) yielded a jump of over 4 percentage points. Extending this to a longer history ($w = 6$) provided another 1.64% gain. Testing further context windows yielded smaller and smaller gains, then losses. This indicates diminishing returns as the conversation history grows in size and eventually negative returns.

This confirms that the DialogueRNN is finding signal from the temporal dynamics in the conversational history. The success at $w = 6$ suggests that longer emotional arcs require a wider field of view for more accurate classification. The longer the context window gets, the more useful history becomes for specific use-cases like escalating arguments, or any sustained emotional shifts; however,

the more specific the use-case gets, accuracy for the general case goes down. This is likely why the accuracy goes down with the longest context windows tested.

Note: some experiments on longest context windows are not recorded in open sourced materials

4.2 Ablation Observations

Ablation studies showed that audio-only (68.22%) and text-only (67.82%) models performed very similarly in isolation. The combined model beat both the isolated models by more than 11 percentage points. This shows high complementarity, the cross-modality is having an effect. The text likely captures *sentiment*, while audio captures *intensity of that sentiment*. Together they stabilize the model. The audio-only model also degraded faster in later epochs, suggesting it is more prone to overfitting than the pre-trained textual representations.

5 Conclusion

This project addressed the "context-blindness" of multimodal transformers. Integrating MemoCMT with DialogueRNN allowed me to combine deeper feature fusion with temporal modeling. The results indicate that including conversational history significantly boosts performance, allowing the model to interpret emotional cues more effectively.

References

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